Fourier Series and PDEs Problem Sheet 8

1. An infinite straight metal rod has a square cross-section whose sides are of length L. The temperature $T(x, y)$ in each cross-section satisfies the boundary value problem given by Laplace's equation

$$\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} = 0 \quad \text{for} \quad 0 < x < L,\ 0 < y < L,$$

with the boundary conditions

$$T(0, y) = 0 \quad \text{for} \quad 0 < y < L, \tag{1}$$

$$T(L, y) = 0 \quad \text{for} \quad 0 < y < L, \tag{2}$$

$$T(x, 0) = 0 \quad \text{for} \quad 0 < x < L, \tag{3}$$

$$T(x, L) = T^* \quad \text{for} \quad 0 < x < L, \tag{4}$$

where T^* is a positive constant.

(a) Use the method of separation of variables and the principle of superposition to derive the general series solution satisfying Laplace's equation and the boundary conditions (1)–(2) given by

$$T(x, y) = \sum_{n=1}^{\infty} \sin\left(\frac{n\pi x}{L}\right)\left(A_n \cosh\left(\frac{n\pi y}{L}\right) + B_n \sinh\left(\frac{n\pi y}{L}\right)\right),$$

where A_n and B_n are constants. Determine the constants A_n and B_n for which the general series solution satisfies the boundary conditions (3)–(4).

[*You may quote the formulae for the Fourier coefficients of a sine series.*]

(b) By considering three similar problems, show that $T = T^*/4$ at the centre of the square.

2. An infinite straight metal rod of constant thermal conductivity k has cross-section that is a path-connected region S bounded by a simple closed curve C. The temperature $T(x, y)$ in each cross-section satisfies Poisson's equation

$$-k\left(\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2}\right) = Q(x, y) \quad \text{for} \quad (x, y) \in S,$$

with Newton's law of cooling giving the boundary condition

$$-k\frac{\partial T}{\partial n} = h\,(T - T_a) \quad \text{for} \quad (x, y) \in C,$$

where Q is the given volumetric heat source, h is the constant heat transfer coefficient, T_a is the constant ambient temperature and $\partial T/\partial n$ denotes the outward normal derivative of T on C.

(a) Use *Green's Theorem* to show that there is at most one solution if $h > 0$, and that if $h = 0$ then any two solutions differ by a constant.

(b) Now take the region S to be the disc of radius a with centre at the origin $(0, 0)$.

(i) Find the cylindrically symmetric solution if Q is constant and $h > 0$.

(ii) Give an example to show that there exists a $h < 0$ for which the solution is not unique if $Q = 0$ and $T_a = 0$.

3. (a) Suppose that the temperature $T(r,\theta)$ satisfies Laplace's equation

$$\frac{\partial^2 T}{\partial r^2}+\frac{1}{r}\frac{\partial T}{\partial r}+\frac{1}{r^2}\frac{\partial^2 T}{\partial \theta^2}=0 \quad \text{for} \quad r>0,$$

where (r,θ) are plane polar coordinates. Use the method of separation of variables and the principle of superposition to derive the general series solution given by

$$T(r,\theta)=A_0+B_0\log(r)+\sum_{n=1}^{\infty}\left(\left(A_n r^n+\frac{B_n}{r^n}\right)\cos(n\theta)+\left(C_n r^n+\frac{D_n}{r^n}\right)\sin(n\theta)\right),$$

where A_n, B_n, C_n and D_n are constants. You should state where you impose the constraint that the solution is periodic in θ with period 2π.

(b) Let a, b, T^* and q^* be positive constants, with $a<b$, and let k be the constant thermal conductivity. Derive the solution $T(r,\theta)$ of Laplace's equation in the region

(i) $r>a$ with the boundary conditions

$$T(a,\theta)=T^*\cos^2\theta \quad \text{and} \quad \lim_{r\to\infty} r\frac{\partial T}{\partial r}(r,\theta)=0 \quad \text{for} \quad -\pi<\theta\leq\pi;$$

(ii) $a<r<b$ with the boundary conditions

$$T(a,\theta)=T^*\cos(\theta) \quad \text{and} \quad -k\frac{\partial T}{\partial r}(b,\theta)=q^* \quad \text{for} \quad -\pi<\theta\leq\pi;$$

(iii) $r<a$ with the boundary conditions

$$T(a,\theta)=\begin{cases} 0 & \text{for } -\pi<\theta\leq 0,\\ T^* & \text{for } 0<\theta\leq\pi.\end{cases}$$

[*You may quote the formulae for the Fourier coefficients of a Fourier series.*]

(c) Find the heat flux $\mathbf{q}\cdot\mathbf{n}$ out of each region in part (b) through the boundary at $r=a$, where the heat flux vector $\mathbf{q}=-k\boldsymbol{\nabla}T$ according to Fourier's Law and in each case $\mathbf{n}$ is the outward pointing unit normal to $r=a$.

[*Before taking the limit $r\to a-$ in case (iii), you may find it helpful to sum the series solution for $T_r(r,\theta)$ by making a suitable choice for z in the identity*

$$\frac{z}{1-z^2}=\sum_{m=0}^{\infty} z^{2m+1},$$

which is valid for $z\in\mathbb{C}$ such that $|z|<1$.]